Chengpeng Wang

Postdoctoral Fellow in Computer Science, Purdue

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RESEARCH INTEREST

Software Engineering, Programming Languages, System Security, with a focus on bridging formal methods and machine learning to enhance the reliability and performance of software systems.

EDUCATION

Hong Kong University of Science and Technology *Aug 2019 - Dec 2023*
Ph.D. in Computer Science and Engineering
Committees: *Charles Zhang (advisor)*, *Xiangyu Zhang*, *Shing-Chi Cheung*, *Shuai Wang*
Dissertation: *Enhancing Reliability and Performance of Data-Centric Systems with Static Analysis*

Tsinghua University *Aug 2016 - Jul 2019*
M.S. in Software Engineering

Tsinghua University *Aug 2012 - Jul 2016*
B.S. in Software Engineering, Minor in Math

ACADEMIC AND INDUSTRY APPOINTMENTS

Now Mar 2024	**Postdoctoral Fellow, West Lafayette, IN, USA** Neuro-symbolic code auditing, working with Prof. Xiangyu Zhang
July 2023 May 2023	**Research Intern at Veridise, Remote** Static bug detection of zero-knowledge proof circuits
Aug 2022 Feb 2021	**Research Intern at Ant Group, Shenzhen, China** Performance optimization of database-backed application
Aug 2019 Feb 2019	**Research Intern at Sourcebrella, Shenzhen, China** Inconsistency detection of build systems

HONORS AND AWARDS

Postdoc Travel Award, Purdue	*2024*
ACM SIGARCH Best Paper Award, ASPLOS'24	*2024*
UGC Research Travel Grant, HKUST	*2022, 2023*
Ant Group Outstanding Collaboration Award	*2023*
ACM SIGPLAN Distinguished Paper Award, OOPSLA'22	*2022*
ACM SIGPLAN PAC Award	*2022*
Outstanding Tutor Award, Student Learning and Development Center, THU	*2019*
Future Academician Scholarship, honored 100 students at Tsinghua University	*2016*
Scholarship for Academic Excellence, Tsinghua University	*2013, 2014*

GRANTS

PI. "*Neuro-Symbolic Static Bug Detection for Reliable Software Systems*". Submitted to ***ARC Discovery Early Career Researcher Award Scheme***.

Co-PI, with Xiangyu Zhang (PI). "*Specification-Aware Context-Directed Testing In the Presence of Library Upgrades*". Submitted to ***Amazon***.

PUBLICATIONS

Highlight: I authored 6 first-author papers and 1 co-first-author paper (labeled with #) published in top-tier conferences and journals, including **ICSE**, **FSE**, **ASE**, **TOSEM**, **OOSPLA**, **NeurIPS**, and **EMNLP**. Additionally, I authored 4 corresponding-author-papers (labeled with $*$) published in **ICSE**, **ASE**, **TOSEM**, and **TSE**. The underlined first authors are Ph.D. students under my mentoring.

PAPERS IN PROGRESS

[P1] Jinyao Guo#, **Chengpeng Wang**#, Xiangzhe Xu, Zian Su, and Xiangyu Zhang. RepoAudit: An Autonomous LLM-Agent for Repository-Level Code Auditing. In **arXiv 2025**. (Under Review)

[P2] **Chengpeng Wang**, Yifei Gao, Wuqi Zhang, Xuwei Liu, Qingkai Shi, and Xiangyu Zhang, LLMSA: A Compositional Neuro-Symbolic Approach to Compilation-free and Customizable Static Analysis, In **arXiv 2024**. (Under Review)

[P3] Chao Wang, Li Lin, **Chengpeng Wang**, Jiafeng Huang, Congxia Wu, and Rongxin, ReachCheck: Compositional Library-Aware Call Graph Reachability Analysis in the IDEs. (Under Review)

[P4] Mingwei Zheng, Danning Xie, Qingkai Shi, **Chengpeng Wang**, and Xiangyu Zhang, Validating Network Protocol Parsers with Traceable RFC Document Interpretation, In **ISSTA 2025**: International Symposium on Software Testing and Analysis. (Major Revision)

[P5] Yuan Li, Peisen Yao, Kan Yu, **Chengpeng Wang**, Yaoyang Ye, Song Li, Meng Luo, Yepang Liu, and Kui Ren. Understanding Industry Perspectives of Static Application Security Testing Evaluation, In **FSE 2024**: International Conference on the Foundations of Software Engineering. (Major Revision)

JOURNAL PUBLICATIONS

[J1] Rongxin Wu, Zhiling Huang, Zige Tian, **Chengpeng Wang***, and Xiangyu Zhang, PackHunter: Recovering Missing Packages for C/C++ Projects, In **TSE**: Transactions on Software Engineering

[J2] Yi Sun, **Chengpeng Wang**, Gang Fan, Qingkai Shi, and Xiangyu Zhang, Fast and Precise Static Null Exception Analysis with Synergistic Preprocessing, In **TSE**: Transactions on Software Engineering

[J3] Wensheng Tang, Dejun Dong, Shijie Li, **Chengpeng Wang***, Peisen Yao, Jinguo Zhou, and Charles Zhang, Octopus: Scaling Value-Flow Analysis via Parallel Collection of Realizable Path Conditions, In **TOSEM**: Transactions on Software Engineering and Methodology

[J4] **Chengpeng Wang**, Wenyang Wang, Peisen Yao, Qingkai Shi, Jinguo Zhou, Xiao Xiao, and and Charles Zhang, Anchor: Fast and Precise Value-Flow Analysis for Containers via Memory Orientation, In **TOSEM**: Transactions on Software Engineering and Methodology

CONFERENCE PUBLICATIONS

[C1] Wei Chen, Bowen Zhang, **Chengpeng Wang**, Wensheng Tang, and Charles Zhang, Seal: Towards Diverse Specification Inference for Linux Interfaces from Security Patches, In **EuroSys 2025**: European Conference on Computer Systems

[C2] **Chengpeng Wang**, Wuqi Zhang, Zian Su, Xiangzhe Xu, Xiaoheng Xie, and Xiangyu Zhang, LLMDFA: Analyzing Dataflow in Code with Large Language Models, In **NeurIPS 2024**: Annual Conference on Neural Information Processing Systems

[C3] **Chengpeng Wang**, Wuqi Zhang, Zian Su, Xiangzhe Xu, and Xiangyu Zhang, Sanitizing Large Language Models in Bug Detection with Data-Flow, In **Findings of EMNLP 2024**: Empirical Methods in Natural Language Processing

[C4] Li Lin, Zongyin Hao, **Chengpeng Wang**, Zhuangda Wang, Rongxin Wu, and Gang Fan, SQLess: Dialect-Agnostic SQL Query Simplification, In **ISSTA 2024**: International Symposium on Software Testing and Analysis

[C5] **Chengpeng Wang**, Jipeng Zhang, Rongxin Wu, and Charles Zhang, DAInfer: Inferring API Aliasing Specifications from Library Documentation via Neurosymbolic Optimization, In **FSE 2024**: International Conference on the Foundations of Software Engineering

[C6] Bowen Zhang, Wei Chen, Peisen Yao, **Chengpeng Wang**, Wensheng Tang, and Charles Zhang, SIRO: Empowering Version Compatibility in Intermediate Representations via Program Synthesis, In **ASPLOS 2024**: ACM Conference on Architectural Support for Programming Languages and Operating Systems

[C7] Hao Ling, Heqing Huang, **Chengpeng Wang**, Yuandao Cai, and Charles Zhang, GiantSan: Efficient Memory Sanitization with Segment Folding, In **ASPLOS 2024**: ACM Conference on Architectural Support for Programming Languages and Operating Systems **(ACM SIGARCH Best Paper Award)**

[C8] Rongxin Wu, Yuxuan He, Jiafeng Huang, **Chengpeng Wang***, Wensheng Tang, Qingkai Shi, Xiao Xiao, and Charles Zhang, LibAlchemy: A Two-Layer Persistent Summary Design for Taming Third-Party Libraries in Static Bug-Finding Systems, In **ICSE 2024**: International Conference on Software Engineering

[C9] Wensheng Tang[#], **Chengpeng Wang**[#], Peisen Yao, Rongxin Wu, Xianjin Fu, Gang Fan, and Charles Zhang, DCLink: Bridging Data Constraint Changes and Implementations in FinTech Systems, In **ASE 2023** : International Conference on Automated Software Engineering

[C10] **Chengpeng Wang**, Peisen Yao, Wensheng Tang, Gang Fan, Charles Zhang, Synthesizing Conjunctive Queries for Code Search, In **ECOOP 2023**: European Conference on Object-Oriented Programming

[C11] Zongyin Hao, Quanfeng Huang, **Chengpeng Wang**, Jianfeng Wang, Yushan Zhang, Rongxin Wu, and Charles Zhang, Pinolo: Detecting Logical Bugs in Database Management Systems with Approximate Query Synthesis, In **ATC 2023**: USENIX Annual Technical Conference

[C12] **Chengpeng Wang**, Gang Fan, Peisen Yao, Fuxiong Pan, and Charles Zhang, Verifying Data Constraint Equivalence in FinTech Systems, In **ICSE 2023**: International Conference on Software Engineering

[C13] Rongxin Wu, Minglei Chen, **Chengpeng Wang***, Gang Fan, Jiguang Qiu, and Charles Zhang, Accelerating Build Dependency Error Detection via Virtual Build, In **ASE 2022** : International Conference on Automated Software Engineering

[C14] **Chengpeng Wang**, Peisen Yao, Wensheng Tang, Qingkai Shi, and Charles Zhang: Complexity-Guided Container Replacement Synthesis, In **OOPSLA 2022**: SIGPLAN Conference on Objected Oriented Programming, Systems, Languages and Applications **(ACM SIGPLAN Distinguished Paper Award)**

[C15] Gang Fan, **Chengpeng Wang**, Rongxin Wu, Xiao Xiao, Qingkai Shi, and Charles Zhang: Escaping Dependency Hell: Finding Build Dependency Errors with the Unified Dependency Graph, In **ISSTA 2020**: International Symposium on Software Testing and Analysis

WORKSHOP and POSTER PUBLICATIONS

[W1] Ming Liang, Xiaoheng Xie, Gehao Zhang, Xunjin Zheng, Peng Di, Wei Jiang, Hongwei Chen, **Chengpeng Wang**, and Gang Fan, RepoGenix: Dual Context-Aided Repository-Level Code Completion with Language Models, In Poster Track, **ASE 2024**: International Conference on Automated Software Engineering

[W2] **Chengpeng Wang**, CodeSpider: Automatic Code Querying with Multi-modal Conjunctive Query Synthesis, In SRC track, **SPLASH 2022**: SIGPLAN conference on Systems, Programming, Languages, and Applications: Software for Humanity, Student Research Competition

[W3] **Chengpeng Wang**, Yixiao Yang, Han Liu, and Le Kang: Statistical API Completion Based on Code Relevance Mining, In **MAINT 2019**: International Workshop on Mining and Analyzing Interaction Histories

ACADEMIC SERVICES

Program Organizing Committee Member:

International Workshop on Language Models and Programming Languages (LMPL)	*2025*

Program Committee Member:

International Conference on Automated Software Engineering (ASE)	*2025*
International Conference on AI Foundation Models and Software Engineering (Forge)	*2024, 2025*
International Conference on the Foundations of Software Engineering (FSE)	*2025*
International Symposium on Software Testing and Analysis (ISSTA)	*2025*
SPLASH, Student Research Contest Track	*2024*
International Conference on Automated Software Engineering (ASE), Industrial Track	*2024*
International Symposium on Software Reliability Engineering (ISSRE)	*2024*

Journal Reviewer:

Transactions on Software Engineering (TSE)	2024

Artifact Evaluation Committee Member:

International Conference on OOP, Systems, Languages, and Applications (OOPSLA)	*2024*
ACM SIGPLAN Conference on Programming Language Design and Implementation (PLDI)	*2023*
International Conference on the Foundations of Software Engineering (FSE)	*2022*
International Symposium on Software Testing and Analysis (ISSTA)	*2022*

Volunteer:

International Conference on OOP, Systems, Languages, and Applications (OOPSLA)	*2022*
Student Volunteer at International Symposium on Software Testing and Analysis (ISSTA)	*2019*
Long-term mentor of SIGPLAN-M	*2024, 2025*

TEACHING

CS 592: AI and Security, Purdue University	*Fall 2024*
COMP 3021: Java Programming, HKUST	*Spring/Fall 2022/2023*
COMP 4631: Computer and Communication Security, HKUST	*Fall 2021*
COMP 3111/H: Software Engineering, HKUST	*Fall 2020*
COMP 2011: Programming with C++, HKUST	*Spring 2020*
Haskell: Functional Language Programming, THU	*Spring 2019*
Automaton and Formal Logic, THU	*Fall 2018*
Drop-in Tutoring for STEM Courses, THU	*Spring/Fall 2018/2019*

ADVISING AND MENTORING

Jinyao Guo (Graduate Student, Purdue University) *Nov 2024 - Present*
Currently a Ph.D. student at Purdue University
Working on LLM-driven code auditing and coauthored a preprint [P1].

Mingwei Zheng (Graduate Student, Purdue University) *Aug 2024 - Present*
Currently a Ph.D. student at Purdue University
Working on network protocol bug detection. Coauthored a paper under major revision [P4].

Yifei Gao (Graduate Student, Purdue University) *May 2024 - Present*
Currently a Ph.D. student at Purdue University
Working on automated translation from C to safer Rust.

Pengxiang Huang (Graduate Student, Northwestern University) *May 2024 - Present*
Currently a Master student at Northwestern University
Working on automated translation from C to safer Rust.

Bowen Zhang (Graduate Student, HKUST) *Oct 2022 - Dec 2023*
Currently a Ph.D. student at Hong Kong University of Science and Technology
Worked on IR Translator Synthesis and coauthored a publication at ASPLOS 2024 [C6].

PRESENTATIONS AND INVITED TALKS

Advances in AI-powered Code Security: Next-Level Bug Detection
Uber's Programming Systems Team. Virtual — Feb 2025
CodeQL@GitHub. Virtual — Jan 2025

Neuro-Symbolic Static Analysis for Reliable Software Systems
ByteDance. Virtual — Jan 2025
Department of Computing, Hong Kong Polytechnic University. Virtual — Jan 2025
Institute of Data Science, The University of Hong Kong. Virtual — Dec 2024

Neuro-Symbolic Static Analysis for Reliable and Performant Software Systems
School of Computer Science, Nanjing University. Nanjing, China — Dec 2024

LLMDFA: Analyzing Dataflow in Code with Large Language Models
Ant Group. Virtual — Dec 2024

When Static Analysis Meets Large Language Models: A Neuro-Symbolic Approach
Boston Computation Club. Virtual — Nov 2024
AI-CyberSecurity Research Lunch Talk, Texas A&M University (TAMU). Virtual — Oct 2024
School of Software, Tsinghua University. Virtual — Oct 2024

Towards Enhancing Reliability and Performance of Data-Centric Systems with Static Analysis
School of Informatics, Xiamen University. Xiamen, China — Aug 2023

Synthesizing Conjunctive Queries for Code Search
ByteDance. Virtual — Jun 2023

Complexity-Guided Container Replacement Synthesis
AST Lab, ETH Zurich. Virtual — Mar 2023